

Video Transcript

Video 8: Creating a Student Grades Database Using Dictionaries (08:59)

So, let's take a look at creating another version of the database. In this case, instead of having the students as a list or an array, we're going to have a dictionary. And the same with Psets and grades. Now, remember with grades, we have the student ID and the Pset ID, and we could have the Pset first and the student ID second. But let's have the student ID first, and then we'll get the Pset ID second. So, that's going to be our structure. So, let's take a look. We need to be able to add students, add Psets, and add grades just like before. So, we'll pass in the database the student ID and the student name. So, in this case, what do we need to do? We need to get hold of first `db[ 'students' ]`, since we're going to add it to the students. And now, it depends what we're going to have here. Let's have the student ID as the key. So, we'll go with `student_id` as the key. So, now we've got hold of the right dictionary, and now we just need the `student_name`. So, that should be it. Let's take a look at `addPset`.

So, with the pset, we're going to need to get hold of the database, but this time we'll get hold of the psets. And we need the `pset_id`. Okay. And that'll be `pset_total_points`. and that will be `pset_total_points`. And now, let's take a look at the `addGrade`. So, again we go to `db[ 'grades' ]`. Now we've got the right dictionary. Now, we said we'd get hold of first the `student_id`. So, now we've got the `student_id` and now there may not be a grade yet in there. So, let's think we need to check first if the `student_id` is in grades. So, if the `student_id` is not in the grades, we need to put the `student_id` there. Okay. And that will be for the `student_id`, then this is going to be empty.

Let's put make this right. We need a block for the if statement, that's correct. Now, if they are in the grades, we can go ahead and get the `student_id`. Okay, we've got that. Now as well, we need to get the `pset_id`. And finally, we can give them the points that they scored. and finally, we can give them the points that they scored. Lets check this. It should be fine. We're not executing anything here. Let's try to add a student. I've got some here, so let's add Alice to the database and see if we're okay. Let's list out the db. Now we need to be careful here. We really need to create the db first. So, let's create an empty db. So, and we need `db=empty()`. Now we've got a database. Let's rerun this. Since we can add... So, now we've added Alice, a student.

Let's try and add... Let's remove Alice from this. And I've already got these typed in, so let's see if we can add the pset grades and the grades here. So, now we've got students, psets, and grades. It looks okay. Let's see if we can now retrieve some of these. Let me give myself some clean space here. So, we've entered the students and the grades. We've got scores, now we need to get the grades. I'm going to write a helper function here, so this is going to help us when we're dealing with dictionaries. So, we're going to pass in a dictionary and a key, and we need to check if the key exists. So, if we do have a key, we want to return that dictionary with a key. Otherwise, we'll print out none. We'll return none. So, instead of throwing an error, this will save us quite a bit. Let's take a look here. So, now let's get a student grade. Let's have a studentGrade. I want to get student grades. Let's see if I can get all the student grades. What do I need to do? Well, I'm going to return, but I need to get. And now, we're going to pass in the dictionary and the key is going to be the student_id. That should be it, because that will return to us the whole dictionary. Let's call db. Let's get student grades for one. We've got to get the grades first from the database, correct? Now we're in students; that's correct. We need to return this. Good, now we've got it. So, this is problem set one, problem set two and problem set three.

Excellent. So, now we've got the student grades. So, why don't you now write the following, studentGradeOn? So, we're going to have the studentid and the problem set ID. So, why don't you write that for for student 1 on problem set 3? Okay, check that out. Okay, bye for now.